Dynamic Evolving Neural-Fuzzy Inference System for Rainfall-Runoff (R-R) Modelling

A. Talei[1], L.H.C. Chua[2] and C. Quek[3]
[1]DHI-NTU Water & Environment Research Centre and Education Hub
Nanyang Technological University
50, Nanyang Avenue, Singapore 639798
SINGAPORE
[2]School of Civil & Environmental Engineering
Nanyang Technological University
50, Nanyang Avenue, Singapore 639798
SINGAPORE
[3]Center for Computational Intelligence, School of Computer Engineering
Nanyang Technological University
50, Nanyang Avenue, Singapore 639798
SINGAPORE
E-mail: atalei@pmail.ntu.edu.sg

Abstract: *Dynamic Evolving Neural-Fuzzy Inference System (DENFIS) is a Takagi-Sugeno-type fuzzy inference system for online learning which can be applied for dynamic time series prediction. To the best of our knowledge, this is the first time that DENFIS has been used for rainfall-runoff (R-R) modeling. DENFIS model results were compared to the results obtained from the physically-based Storm Water Management Model (SWMM) and an Adaptive Network-based Fuzzy Inference System (ANFIS) which employs offline learning. Data from a small (5.6 km^2) catchment in Singapore, comprising 11 separated storm events were analyzed. Rainfall was the only input used for the DENFIS and ANFIS models and the output was discharge at the present time. It is concluded that DENFIS results are better or at least comparable to SWMM, but similar to ANFIS. These results indicate a strong potential for DENFIS to be used in R-R modeling.*

Keywords: *Rainfall-Runoff Modelling, Neuro-Fuzzy, DENFIS.*

1. INTRODUCTION

The rainfall-runoff (R-R) process can be considered as one of the most important problems in hydrology. To-date many approaches and methods have been introduced to model this process. These methods can be categorized into two main groups: (1.) Physically-based models such as the Storm Water Management Model (SWMM) and Sacramento Soil Moisture Accounting model (SAC-SMA); and (2.) System theoretic models such as linear and non-linear regression models, Autoregressive Moving Average with Exogenous Inputs (ARMAX), Artificial Neural Networks (ANN), and Neuro-Fuzzy Systems (NFS).

Neuro-Fuzzy Systems implement fuzzy inference which is a process that maps a given input to an output using fuzzy logic. A fuzzy inference system (FIS) or fuzzy rule based system in general includes four main components: (1.) a fuzzification interface which transforms the crisp values into fuzzy values based on the degrees of matching with linguistic variables, (2.) an interface engine which performs inference operations on the rules, (3.) a rule base containing fuzzy IF-THEN rules, and (4.) a defuzzification interface which transforms the fuzzy results back into a crisp output. Based on the method chosen for determining its output, FIS can be categorized as either linguistic or precise models. One of the most frequently used precise fuzzy models is the Takagi-Sugeno FIS (Takagi & Sugeno, 1985). In the Takagi-Sugeno FIS, a fuzzy rule is constituted by a weighted linear combination of crisp inputs rather than a fuzzy set. The Dynamic Evolving Neural-Fuzzy Inference System or DENFIS (Kasabov & Song, 2002) uses Takagi-Sugeno type fuzzy inference engine for online learning which can be used in dynamic time series prediction.

In this study, the capabilities of DENFIS have been assessed for an event-based R-R modelling study for a sub-catchment (5.6 km^2) in Kranji, Singapore. The results obtained from DENFIS were compared with results obtained from a R-R model developed using an Adaptive Network-based Fuzzy Inference System or ANFIS (Jang, 1993) and a deterministic model (SWMM). Learning in ANFIS is offline and can be considered as global learning on the input space. Therefore, the capabilities of DENFIS with online local learning can be compared to ANFIS which uses offline global learning methodology. The input used for the DENFIS and ANFIS models is rainfall and the output is discharge. This is consistent with the deterministic modelling approach where both rainfall and discharge are used to calibrate the model, but only rainfall is used as input at the model prediction stage.

2. DYNAMIC EVOLVING NEURAL-FUZZY INFERENCE SYSTEM (DENFIS)

DENFIS modelling evolves through incremental learning resulting in local element adjustment as new data is introduced to the model in sequence (Kasabov & Song, 2002). Local element learning is implemented via a scatter partitioning of the input space for the creation of fuzzy inference rules. DENFIS utilizes an evolving, online clustering method called the Evolving Clustering Method (ECM). ECM is a distance-based, fast, one-pass algorithm used for the dynamic estimation of the number of clusters in a data set and also allocates the position of cluster centres in the input data space. In this clustering method, the maximum distance between a data point and the cluster centre must be less than a predefined threshold value, *Dthr*, (a clustering parameter); thus, *Dthr* controls the size and hence the number of clusters to be created. *Dthr* = 0.1 gave the best model performance for our study and this value was adopted.

The output of DENFIS includes *n* fuzzy rules of which the j^{th} rule is shown in Eq. (1):

If x_1 is R_{j1} and x_2 is R_{j2} and ... and x_m is R_{jm}, then y is $f_j(x_1, x_2, ..., x_m)$ (1)

where "x_i is R_{ji}" (i = 1, 2, ...m; j = 1, 2, ...n) are $n \times m$ fuzzy propositions as *n* antecedents form *n* fuzzy rules; x_i (i = 1, 2, ...m) are antecedent variables defined over the universes of discourse X_i (i = 1, 2, ...m); R_{ji} (i = 1, 2, ...m; j = 1, 2, ...n) are fuzzy sets defined by their fuzzy membership functions given by $\mu_{Rji} : X_i \rightarrow [0,1]$; and *y* is the consequent. The membership functions $\mu(x)$ used in DENFIS models is of the triangular type with three parameters as given by the following equation:

$$\mu(x) = MF(x,a,b,c) = \begin{cases} 0 & x \le a \\ \dfrac{x-a}{b-a} & a \le x \le b \\ \dfrac{c-x}{c-b} & b \le x \le c \\ 0 & c \le x \end{cases} \tag{2}$$

where *b* is the cluster centre; $a = b - d \times Dthr$; $c = b + d \times Dthr$; and *d* varies between 1.2 to 2.

In the consequent part of Eq. (1), f_j is a linear function. In the DENFIS model, f_j is determined by the linear least square estimator (LSE) method. Each of the linear functions can be expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_m x_m \tag{3}$$

where $\beta_0, \beta_1, \beta_2$, and β_m are parameters to be estimated. Using *p* learning data pairs, the parameters are estimated by the weighted recursive least square estimator which employs a forgetting factor, λ which has typical values between 0.8 to 1 (Kasabov & Song, 2002). In DENFIS, rules are created and updated at the same time with the partitioning of the input space by the ECM; presenting new data pairs to the model may result in the creation of new rules or updating of existing rules. In general, a new rule is created if a new cluster centre is found by ECM and a rule is updated if another data point is added to an existing cluster.

3. METHODOLOGY

3.1. Study Site and Data Used

The total area of the catchment under study is about 5.6km^2, and the land use consists of 33% (about 1.8km^2) high-density residential areas with the remainder consisting of undeveloped land areas which are mainly covered by vegetation. The runoff in the study site is served mainly by a concrete-lined drainage system. CP1 (Figure 1) is the final discharge point of the catchment. Discharge and rainfall measurements recorded at CP1 were used in this study. The time series of rainfall and discharge measured at 5-min intervals from 16 Dec 2004 to 3 Nov 2006 were selected for this study. A total of 47 rainfall events was selected from the original recorded time series and analyzed for this study. Using correlation analysis between rainfall and direct runoff volume, Tan *et al.* (2008) selected 10 events for calibrating a SWMM model. Therefore, event-based R-R modeling is considered here for ease of comparison with the SWMM model; since SWMM provides results for direct runoff only. Thus, the measured rainfall and direct runoff component of the hydrographs separated for this purpose were used in our study.

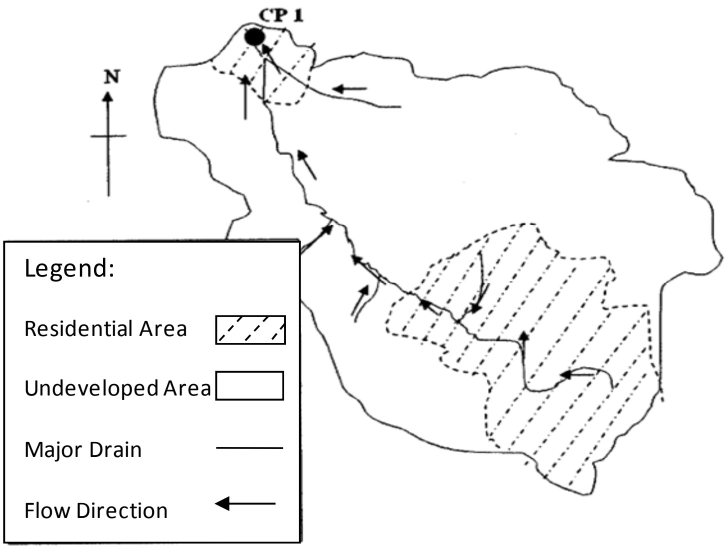

Figure 1 Schematic map of the study site.

3.2. Model Verification

Four error statistics were used to evaluate the performance of the DENFIS model. The root mean square error or *RMSE* (m^{3}sec^{-1}) is defined as:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(Q_i - \hat{Q}_i)^2}{n}} \tag{4}$$

where Q_i and $\hat{Q}_i$ are the observed and simulated discharges for the i^{th} observation, respectively; and *n* is the total number of observations. *RMSE* accords extra importance of the outliers in the data set and hence is biased towards errors in the simulation of high flow rates (Dawson *et al.*, 2006). The Nash-Sutcliffe coefficient of efficiency or *CE* (Nash & Sutcliffe, 1970) is defined as:

$$CE = 1 - \frac{\sum_{i=1}^{n} (Q_i - \hat{Q}_i)^2}{\sum_{i=1}^{n} (Q_i - \overline{Q})^2} \tag{5}$$

where $\overline{Q}$ is the average value of the observed flows. *CE* ranges between 1 (perfect fit) and $-\infty$ and null or negative *CE* values indicate that the mean value of the observed time series could be a better predictor than the model (Krause *et al.*, 2005). *CE* has been found to be sensitive to extreme values (Abrahart et al., 2004). The coefficient of determination or r^2 is defined as follows:

$$r^2 = \left[\frac{\sum_{i=1}^{n} (Q_i - \overline{Q})(\hat{Q}_i - \tilde{Q})}{\sum_{i=1}^{n} (Q_i - \overline{Q})^2 \quad \sum_{i=1}^{n} (\hat{Q}_i - \tilde{Q})^2} \right]^2 \tag{6}$$

where $\tilde{Q}$ is the average value of the simulated flows. r^2 shows the degree of co-linearity between the observed and simulated time series and ranges from 0.0 to 1.0, with higher values indicating higher degree of co-linearity. r^2 is sensitive to outliers and insensitive to additive and proportional differences between modeled and observed data (Abrahart *et al.*, 2004). In addition to the overall goodness-of-fit, accurate prediction of peak flow is important. Thus, the Relative Peak Error or *PE* was included in this study to evaluate the ability of the proposed models to accurately predict peak flows. *PE* is defined as:

$$PE = \frac{\left| (Q_p) - (\hat{Q}_p) \right|}{(Q_p)} \tag{7}$$

where Q_p and $\hat{Q}_p$ are observed and simulated peak flow rates. Values of *PE* closer to zero indicate better estimation of peak flows.

3.3. Model Development and Input Data Selection

In event-based R-R modelling, the training process can be sensitive to the events selected for training the model since characteristics such as the shape (Talei *et al.*, 2010) and lag between the hyetograph and hydrograph can influence model results. In general, a lag time (which may differ from event to event) exists between a rainfall event and the resultant runoff. We use cross-correlation analysis between the time series of the recorded rainfall and discharge to determine the lag. Figure 2 illustrates the distribution of lag times for the recorded events, which can be categorized into four different groups G1, G2, G3, and G4 with lag times of 25, 30, 35, and 40 minutes, respectively. The average and median lags are 32 and 30 min, respectively.

DENFIS and ANFIS models were developed for each of the 4 groups of events; however, only DENFIS and ANFIS model results for G2 (events with 30 min lag) are reported here. G2 is chosen since the lag corresponds to the average and median lags of the recorded events. Among the eleven events in G2, three events (19 Jul 05, 30 Jul 05, 24 Dec 05) were chosen for training and the remainder for testing. The three selected training events had moderate, single peak discharges; in order to avoid training the model using extreme events. Figure 3 shows the hyetographs and hydrographs of the selected training events.

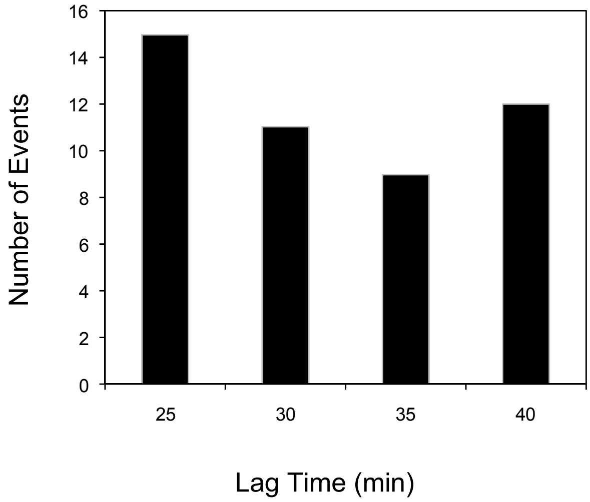

Figure 2 Distribution of lag time for the recorded events.

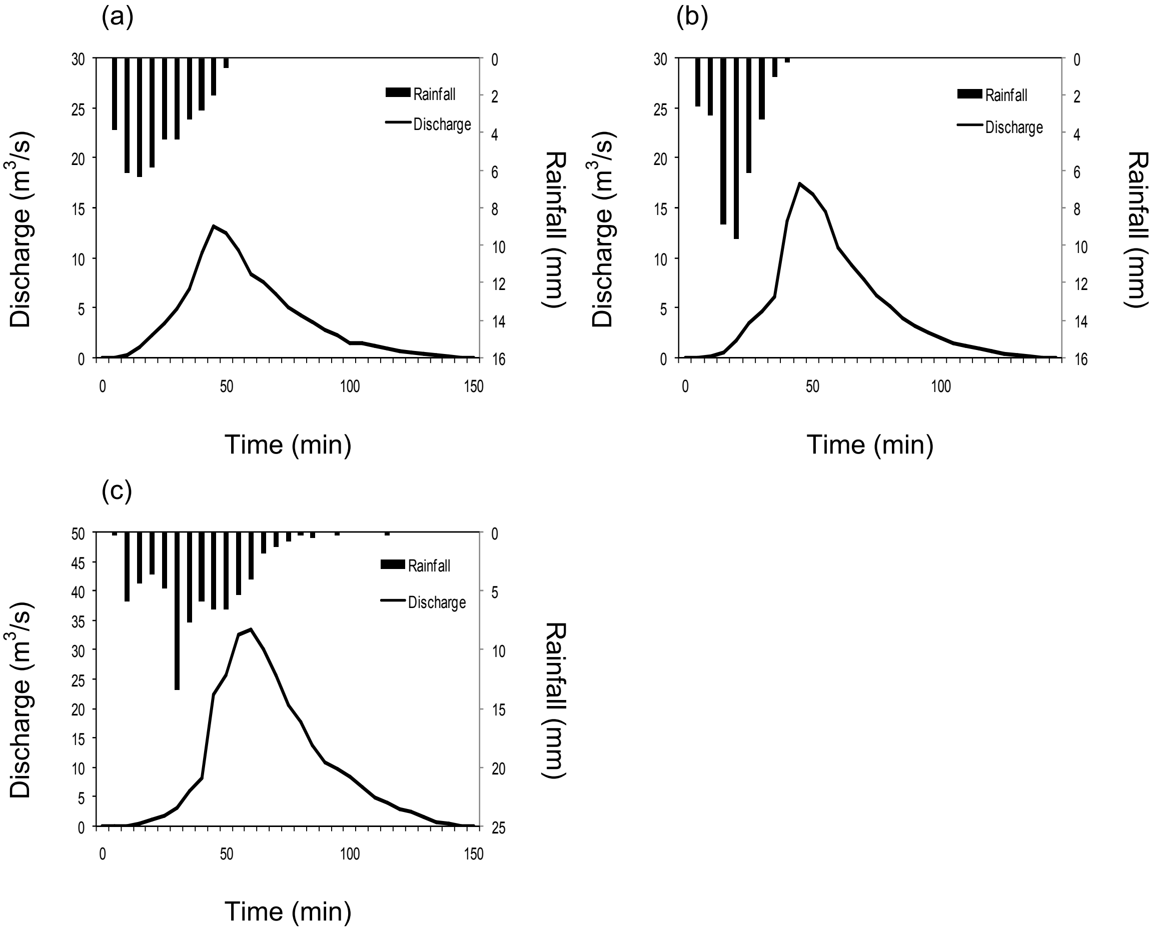

Figure 3 Hydrograph and hyetograph for training events (a) 19 Jul 05, (b) 30 Jul 05, and (c) 24 Dec 05.

Having selected the training events, the number of rainfall inputs was next found by trial-and-error. Our initial analysis showed that using 3 inputs was sufficient for training the DENFIS and ANFIS models. Using a lesser number of inputs would decrease the training performance due presumably to insufficient information while using more than three inputs was found to reduce the training performance due presumably to redundancy in the number of inputs. The best three antecedent rainfall inputs found by trial-and-error for DENFIS, were R(t-3), R(t-4), and R(t-7).

4. RESULTS AND DISCUSSIONS

Online training of the DENFIS model was carried out as mentioned above and the trained DENFIS model was then used to simulate the runoff at time *t*, Q(t) for the 8 remaining events in G2. *Dthr* was fixed at 0.1 and the model performance in simulating Q(t) for all 8 events was evaluated in terms of the error statistics *CE*, r^2, *RMSE*, and *PE*. In order to compare the capabilities of DENFIS in R-R modelling, DENFIS model results were compared with the results obtained by SWMM based on a study by Tan *et al*., (2008) and an ANFIS model which was developed with the same training set and identical number of inputs used for training the DENFIS model. ANFIS is constructed as a five layered Multi Layered Perceptron (MLP) network, and is a Takagi-Sugeno type neuro-fuzzy system that employs back-propagation algorithm to modify the initially chosen membership functions. ANFIS utilizes the least square algorithm to determine the coefficients of the linear output functions (Jang, 1993). The type and number of membership functions allocated to each input should be chosen for the ANFIS model. In this study, two triangular membership functions were used for each input to develop the ANFIS model. ANFIS implemented in the Fuzzy Logic Toolbox, MATLAB (MATLAB, 2008) was used in this study.

Table 1 summarizes the average values of *CE*, r^2, *RMSE*, and *PE* and Figures 4a, 4b, 4c, and 4d show the box plots comparing the testing results by DENFIS, SWMM, and ANFIS models. The box plot shows the distribution of maximum, third quartile, median, first quartile, and minimum values of the error statistics obtained. Outliers are defined as values exceeding 1.5 times the inter-quartile range (difference between the third and the first quartile) (Figure 4c) and are represented by a symbol.

Table 1: Comparison of average *CE*, r^2, *RMSE*, and *PE* values for DENFIS, ANFIS and SWMM models.

Model	*CE*	r^2	*RMSE* (m^3/s)	*PE*
DENFIS	0.791	0.830	1.818	0.132
ANFIS	0.820	0.884	1.525	0.162
SWMM	0.688	0.774	2.227	0.279

As shown, DENFIS generally gives a better fit between the modeled and measured data when compared to SWMM; median *CE* and r^2 values obtained from the DENFIS model (*CE* = 0.823, r^2 = 0.885) are higher than results obtained from the SWMM model (*CE* = 0.719, r^2 = 0.760). In fact, 4 of the 8 events tested by DENFIS have *CE* > 0.9, which is considered a satisfactory fit (Shamseldin, 1997) compared to SWMM which predicts 2 events with *CE* > 0.9. For *PE* (Figure 4d), DENFIS model results are clearly superior with a median *PE* value of 0.059 compared to 0.297 for SWMM. DENFIS model results also gave a significantly lower median value of *RMSE* (Figure 4c) compared to SWMM. This can be attributed to the superiority of DENFIS model in estimating peak flows compared to SWMM (Figure 4d), since *RMSE* is known to be biased towards the simulation of high flow rates (Dawson et al., 2006). SWMM is a popular model and is widely established as a R-R modeling tool. SWMM has been used in this study as a standard against which the DENFIS model may be compared. Based on this brief analysis, our comparisons show that DENFIS model results are comparable, if not better than SWMM, at least for the limited application attempted in this study. This shows the potential of DENFIS for R-R modeling.

As mentioned previously, one of the objectives of this study was to ascertain if improvements could be attained from models employing local learning strategies against models that employ offline learning. The use of models that employ offline learning is popular in data-driven model applications in hydrologic modeling, with the Artificial Neural Network (ANN) and ANFIS being the most widely used. While the use of local learning models is common in other engineering applications, their use in hydrological modeling has not been tested. Comparing between the error statistics obtained by the ANFIS and DENFIS models, it is observed that the ANFIS model is marginally superior compared to DENFIS when evaluated based on *CE*, r^2, and *RMSE* (Figures 4a, 4b, and 4c), while the DENFIS model is found to provide marginally better results compared to ANFIS in estimating the peak discharge (Figure 4d). Thus, our study shows that DENFIS and ANFIS models give mixed or at best, similar results. This observation is in agreement with the study by Hong & White, (2009) that used a

Dynamic Neuro-Fuzzy Local Modelling System (DNFLMS) for flow forecasting in Waikoropupu Springs, New Zealand. DNFLMS is a dynamic Takagi-Sugeno type fuzzy inference system which employs an online and local learning algorithm. The authors compared their results against an ANFIS model and found that both ANFIS and DNFLMS results were comparable. The use of local learning models is an attractive prospect as, intuitively, one would expect better model performance when trained based on a local learning paradigm. However, based on this limited application, DENFIS model results are not better than the results obtained from ANFIS. The reasons for this are currently under investigation and further studies are necessary.

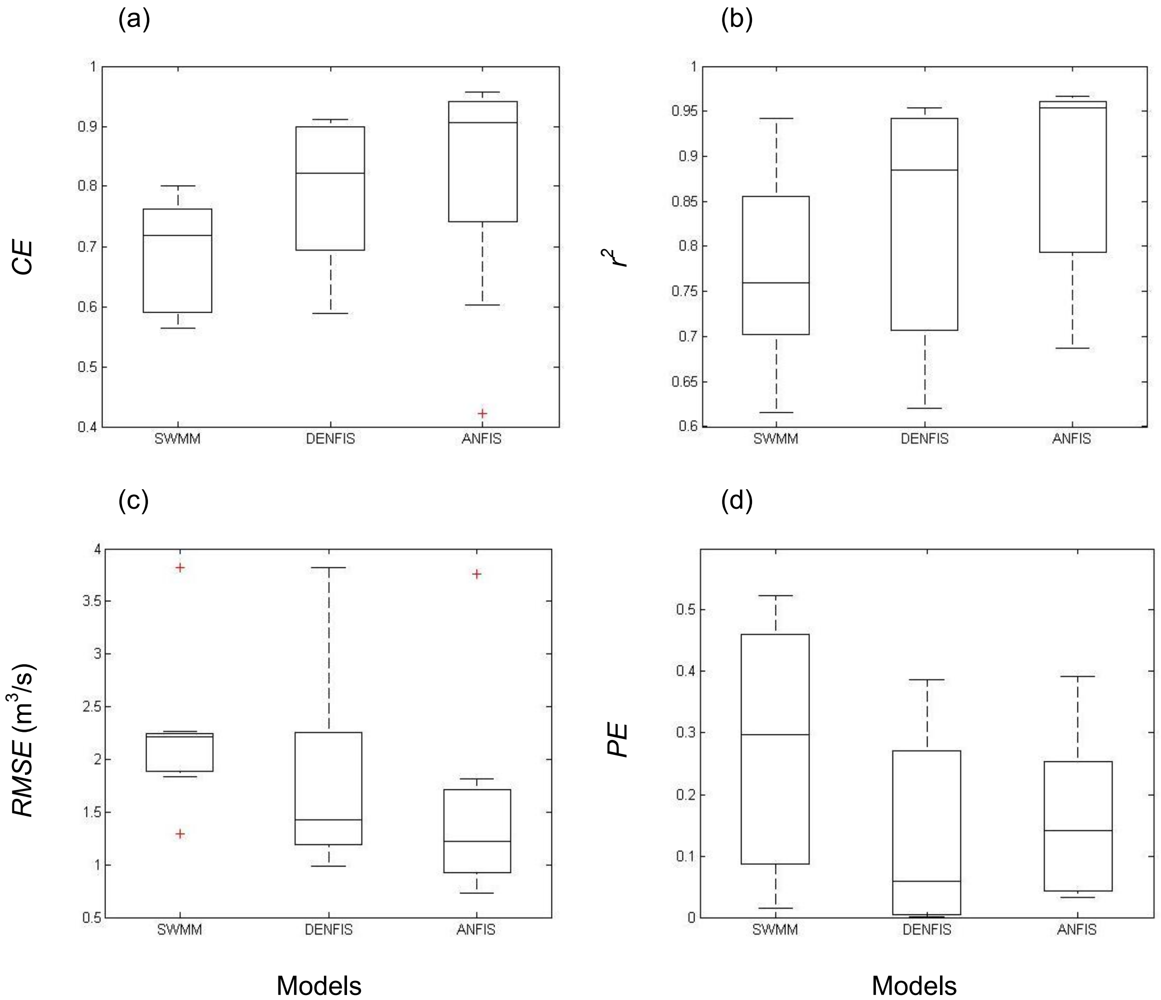

Figure 4 Different models performance in terms of: (a) *CE*, (b) r^2, (c) *RMSE*, and (d) *PE*.

5. CONCLUSIONS

(i) Dynamic Evolving Neural-Fuzzy Inference System (DENFIS) was found to be a superior or at least comparable model compared to SWMM and demonstrates the potential of DENFIS for R-R modelling.

(ii) DENFIS was found to be better at peak estimation compared to SWMM and ANFIS models.

(iii) The local or online learning algorithm used in DENFIS did not show any significant improvement against the global offline learning algorithm used in ANFIS. Further investigations are necessary.

6. REFERENCES

Abrahart, R.J., Kneale, P.E., and See, L.M., (2004), *Neural networks for hydrological modelling*, Taylor & Francis.

Dawson, C.W., See, L.M., Abrahart, R.J., and Heppenstall, A.J., (2006), *Symbiotic adaptive neuro-evolution applied to rainfall-runoff modelling in northern England,* Neural Networks, v. 19, p. 236-247.

Hong, Y.S.T., and White, P.A., (2009), *Hydrological modeling using a dynamic neuro-fuzzy system with on-line and local learning algorithm,* Advances in Water Resources, v. 32, p. 110-119.

Jang, J.-S.R., (1993), *ANFIS: adaptive-network-based fuzzy inference system,* IEEE Transactions on Systems, Man and Cybernetics, v. 23, p. 665-685.

Kasabov, N.K., and Song, Q., (2002), *DENFIS: Dynamic evolving neural-fuzzy inference system and its application for time-series prediction,* IEEE Transactions on Fuzzy Systems, v. 10, p. 144-154.

Krause, P., Boyle, D.P., and Bäse, F., (2005), *Comparison of different efficiency criteria for hydrological model assessment*, Advances in Geosciences, v. 5, p. 89-97.

MATLAB, (2008), *MATLAB Fuzzy Logic Toolbox Reference Manual,* Version 7.6.0
The Math Works Inc. Natick, Mass.

Nash, J.E., and Sutcliffe, J.V., (1970), *River flow forecasting through conceptual models part I - A discussion of principles,* Journal of Hydrology, v. 10, p. 282-290.

Shamseldin, A.Y., (1997), *Application of a neural network technique to rainfall-runoff modelling,* Journal of Hydrology, v. 199, p. 272-294.

Takagi, T., and Sugeno, M., (1985), *Fuzzy identification of systems and its applications to modeling and control,* IEEE Transactions on Systems, Man and Cybernetics, v. 15, p. 116-132.

Talei, A., Chua, L.H.C., and Quek, C., (2010), *A novel application of a neuro-fuzzy computational technique in event-based rainfall-runoff modeling,* Expert Systems with Applications, v. 37, p. 7456-7468.

Tan, S.B.K., Chua, L.H.C., Shuy, E.B., Lo, E.Y.M., and Lim, L.M., (2008), *Performances of rainfall-runoff models calibrated over single and continuous storm flow events*, Journal of Hydrologic Engineering, v. 13, p. 597-607.